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Crack your Interview

on LangChain

Ace your Langchain Interview and become job ready
in 10 minutes



Chains



Agents



RAG



Memory



Tools



Section 1

LangChain Fundamentals

Questions 1-10



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What is LangChain and why do we need it?

LangChain is a Python framework that makes it easy to build applications using Large Language Models (LLMs). Think of it like Lego blocks for AI apps!

WHY WE NEED IT

Without LangChain, you'd have to write tons of code to connect your AI to databases, APIs, and other tools. LangChain does the heavy lifting for you.



Standardizes how you work with different AI models (OpenAI, Claude, etc.)



Connects AI to external data sources and tools



Simplifies building chatbots, RAG apps, and AI agents



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What are the core components of LangChain?

LangChain has **5 main building blocks** that work together:

 **Models:** The AI brain (GPT-4, Claude, etc.) that generates text

 **Prompts:** Templates that tell the AI what to do

 **Chains:** Connect multiple steps together (like a recipe)

 **Agents:** AI that can decide which tools to use

 **Memory:** Remembers past conversations



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How do you initialize a Chat Model in LangChain?

Use `init_chat_model()` to create a connection to any AI model.
It's the easiest way to start!

PYTHON

```
from langchain.chat_models import init_chat_model

# Initialize OpenAI GPT-4
model = init_chat_model("gpt-4o")

# Or use Claude
model = init_chat_model("claude-sonnet-4-5-20250929")

# Send a message
response = model.invoke("Hello, how are you?")
print(response.content)
```

KEY POINT

`init_chat_model()` automatically detects which provider to use based on the model name!



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What is LCEL (LangChain Expression Language)?

LCEL is LangChain's way of connecting components using the **pipe operator (|)**. Think of it like connecting water pipes!

PYTHON

```
from langchain_core.prompts import ChatPromptTemplate
from langchain.chat_models import init_chat_model

# Create a prompt template
prompt = ChatPromptTemplate.from_template(
    "Translate this to French: {text}"
)

# Create the model
model = init_chat_model("gpt-4o")

# Chain them with pipe!
chain = prompt | model

# Run the chain
result = chain.invoke({"text": "Hello world"})
```

SIMPLE ANALOGY

Data flows through the pipe: Input → Prompt → Model → Output



What are Prompt Templates in LangChain?

Prompt Templates are reusable blueprints for creating prompts. Instead of hardcoding text, you use **placeholders** that get filled in later.

PYTHON

```
from langchain_core.prompts import ChatPromptTemplate

# Create a template with variables
template = ChatPromptTemplate.from_messages([
    ("system", "You are a {role} assistant."),
    ("user", "{question}")
])

# Fill in the blanks
prompt = template.invoke({
    "role": "helpful coding",
    "question": "How do I read a file?"
})
```

WHY USE TEMPLATES?

Templates make your code cleaner and let you reuse the same prompt structure with different inputs!



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What are Output Parsers and why use them?

Output Parsers convert the AI's text response into structured data like JSON, lists, or Python objects. Super useful when you need clean data!

PYTHON

```
from langchain_core.output_parsers import JsonOutputParser
from pydantic import BaseModel
```

```
# Define what you want back
```

```
class Movie(BaseModel):
```

```
    title: str
```

```
    year: int
```

```
parser = JsonOutputParser(pydantic_object=Movie)
```

```
# Chain it: prompt | model | parser
```

```
chain = prompt | model | parser
```

```
# Get structured data!
```

```
result = chain.invoke({"query": "Inception"})
```

```
# Returns: {"title": "Inception", "year": 2010}
```



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What are Document Loaders in LangChain?

Document Loaders help you load data from different sources (PDFs, websites, databases) into LangChain. They're the "import" button for your AI!



TextLoader: Load .txt files



PyPDFLoader: Load PDF documents



WebBaseLoader: Scrape websites



CSVLoader: Load CSV files

```
from langchain_community.document_loaders import PyPDFLoader
```

PYTHON

```
loader = PyPDFLoader("my_document.pdf")  
docs = loader.load() # Returns list of Documents
```



What are Text Splitters and why are they needed?

Text Splitters break large documents into smaller chunks. Why? Because AI models have **token limits** - they can only process so much text at once!

PYTHON

```
from langchain_text_splitters import  
RecursiveCharacterTextSplitter  
  
splitter = RecursiveCharacterTextSplitter(  
    chunk_size=1000,      # Max characters per chunk  
    chunk_overlap=200     # Overlap between chunks  
)  
  
chunks = splitter.split_documents(docs)
```

WHY OVERLAP?

Overlap ensures context isn't lost at chunk boundaries. Imagine cutting a book into pages - overlap keeps the story connected!



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How do you handle conversation history?

LangChain uses **message objects** to track conversation history. This helps the AI remember what you've been talking about!

PYTHON

```
from langchain.messages import HumanMessage, AIMessage

# Build conversation history
conversation = [
    HumanMessage("My name is John"),
    AIMessage("Nice to meet you, John!"),
    HumanMessage("What's my name?")
]

# AI will remember: "Your name is John"
response = model.invoke(conversation)
```

MESSAGE TYPES

HumanMessage = user input, AIMessage = model response,
SystemMessage = instructions to the AI



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What is the difference between `invoke()`, `stream()`, and `batch()`?

These are 3 ways to run your LangChain code:



`invoke()`: Run once, wait for complete result



`stream()`: Get results token-by-token (like ChatGPT typing)



`batch()`: Run multiple inputs at once (faster!)

PYTHON

```
# Single call
result = chain.invoke({"text": "Hello"})

# Streaming (for chat UX)
for chunk in chain.stream({"text": "Hello"}):
    print(chunk, end="")

# Batch processing
results = chain.batch([{"text": "Hi"}, {"text": "Bye"}])
```



Section 2

Agents, Tools & RAG

Questions 11-20



What are Agents in LangChain?

Agents are AI systems that can **decide which tools to use** based on the task. Unlike chains (fixed steps), agents choose their own path!

SIMPLE ANALOGY

A chain is like following a recipe. An agent is like a chef who decides what to cook based on available ingredients!

PYTHON

```
from langchain.agents import create_agent

agent = create_agent(
    model="gpt-4o",
    tools=[search_tool, calculator_tool],
    system_prompt="You are helpful assistant"
)

# Agent decides which tool to use!
result = agent.invoke({
    "messages": [{"role": "user", "content": "What's 25 * 4?"}]
})
```



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How do you create custom Tools in LangChain?

Use the **@tool decorator** to turn any Python function into a tool that agents can use!

PYTHON

```
from langchain.tools import tool

@tool
def get_weather(city: str) → str:
    """Get weather for a city."""
    # Your logic here
    return f"It's sunny in {city}!"

@tool
def calculate(expression: str) → str:
    """Calculate a math expression."""
    return str(eval(expression))

# Use in agent
agent = create_agent(
    model="gpt-4o",
    tools=[get_weather, calculate]
)
```

IMPORTANT

The docstring tells the AI when to use the tool. Make it clear and



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What is RAG (Retrieval Augmented Generation)?

RAG = Teaching AI with your own data! Instead of relying only on training data, RAG **retrieves relevant info** from your documents first, then generates an answer.

THE RAG PROCESS

1. User asks question → 2. Find relevant docs → 3. Send docs + question to AI → 4. AI answers using YOUR data



Without RAG: "Who is the CEO?" → AI guesses or says outdated info



With RAG: "Who is the CEO?" → Searches your company docs
→ Gives accurate answer



What are Vector Stores and Embeddings?

Embeddings convert text into numbers (vectors). **Vector Stores** save and search these vectors to find similar content.

SIMPLE ANALOGY

Embeddings = Converting books to GPS coordinates. Vector Store
= Finding the nearest books to your location!

PYTHON

```
from langchain_openai import OpenAIEmbeddings
from langchain_qdrant import QdrantVectorStore

# Create embeddings
embeddings = OpenAIEmbeddings()

# Store documents
vector_store = QdrantVectorStore.from_documents(
    documents=docs,
    embedding=embeddings
)

# Search for similar content
results = vector_store.similarity_search("your query")
```



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How do you implement a basic RAG pipeline?

Here's a complete RAG pipeline in 4 steps:

PYTHON

```
# 1. Load documents
from langchain_community.document_loaders import PyPDFLoader
docs = PyPDFLoader("data.pdf").load()

# 2. Split into chunks
from langchain_text_splitters import RecursiveCharacterTextSplitter
chunks = RecursiveCharacterTextSplitter(chunk_size=1000).split_documents(docs)

# 3. Store in vector store
vector_store.add_documents(chunks)

# 4. Create retrieval tool
@tool
def retrieve(query: str):
    """Search knowledge base."""
    return vector_store.similarity_search(query)
```



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What is Memory in LangChain?

Memory allows AI to remember past conversations. Without memory, each message is like talking to someone with amnesia!



Short-term: Current conversation (stored in messages)



Long-term: Stored in databases (vector stores)

PYTHON

```
from langgraph.checkpoint.memory import InMemorySaver

# Create memory
checkpointer = InMemorySaver()

# Agent with memory
agent = create_agent(
    model="gpt-4o",
    tools=[...],
    checkpointer=checkpointer # Adds memory!
)
```




What is the `create_agent()` function?

`create_agent()` is the new standard way to build agents in LangChain v1. It has built-in streaming, memory, and middleware support!

PYTHON

```
from langchain.agents import create_agent

def send_email(to: str, body: str):
    """Send an email"""
    return f>Email sent to {to}"

agent = create_agent(
    model="gpt-4o",
    tools=[send_email],
    system_prompt="You are an email assistant."
)

result = agent.invoke({
    "messages": [{"role": "user", "content": "Send hi"}]
})
```



How does tool calling work in LangChain?

When you ask a question, the AI **decides if it needs a tool**. If yes, it generates a "tool call" with the function name and arguments.

THE FLOW

1. User asks → 2. AI picks tool → 3. Tool executes → 4. Result sent back → 5. AI generates final answer

```
# User: "What's the weather in Tokyo?"
```

PYTHON

```
# AI generates tool call:
tool_call = {
    "name": "get_weather",
    "args": {"city": "Tokyo"}
}
```

```
# Tool executes and returns result
# AI uses result to answer: "It's 22°C in Tokyo"
```



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What is the ReAct pattern?

ReAct = Reasoning + Acting. The AI thinks out loud before taking action. This makes it smarter and easier to debug!

```
# Question: "Population of France × 2?"
```

PYTHON

Thought: I need to find France's population first

Action: search("France population")

Observation: 67 million

Thought: Now multiply by 2

Action: calculate("67 * 2")

Observation: 134

Thought: I have the answer

Final Answer: 134 million

WHY IT MATTERS

You can see HOW the AI reached its answer, making it easier to fix mistakes!



How do you use `bind_tools()` with models?

`bind_tools()` tells a model which tools it can use. The model then knows how to call them!

```
from pydantic import BaseModel, Field

# Define tool as Pydantic model
class GetWeather(BaseModel):
    """Get weather for a location"""
    location: str = Field(description="City name")

# Bind tools to model
model = init_chat_model("gpt-4o")
model_with_tools = model.bind_tools([GetWeather])

# Model can now use the tool!
response = model_with_tools.invoke("Weather in Paris?")
print(response.tool_calls) # Shows tool call
```

PYTHON



Section 3

Advanced Topics

Questions 21-30



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What is LangGraph and when should you use it?

LangGraph is LangChain's framework for building complex, stateful AI workflows. Use it when simple chains aren't enough!



Use LangChain: Simple tasks, basic chains, quick prototypes



Use LangGraph: Complex workflows, multi-agent systems, human-in-loop

KEY FEATURES

Graphs with nodes and edges, state persistence, streaming, cycles and loops, human approval steps



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How do you implement multi-agent systems?

Multi-agent systems = Multiple specialized agents working together. One agent can call another agent as a tool!

PYTHON

```
# Create specialized agents
researcher = create_agent(model, tools=[search])
writer = create_agent(model, tools=[write_doc])

# Wrap as tool for main agent
@tool
def do_research(query: str):
    """Research a topic"""
    result = researcher.invoke({...})
    return result["messages"][-1].content

# Main agent coordinates others
supervisor = create_agent(
    model, tools=[do_research, call_writer]
)
```



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What is Middleware in LangChain agents?

Middleware lets you intercept and modify agent behavior. Use cases: logging, rate limiting, safety filters, dynamic prompts.

PYTHON

```
from langchain.agents.middleware import dynamic_prompt
```

```
@dynamic_prompt
```

```
def custom_prompt(request):
```

```
    """Change prompt based on context"""
```

```
    role = request.runtime.context.get("role")
```

```
    if role == "admin":
```

```
        return "You have full access."
```

```
    return "You are a helpful assistant."
```

```
agent = create_agent(model, middleware=[custom_prompt])
```




How do you handle streaming in LangChain?

Streaming shows output token-by-token (like ChatGPT typing).
Great for user experience!

PYTHON

```
# Stream from a chain
for chunk in chain.stream({"input": "Hello"}):
    print(chunk, end="", flush=True)

# Stream from an agent with modes
for event in agent.stream(
    {"messages": [...]},
    stream_mode=["updates", "custom"]
):
    print(event)
```

STREAM MODES

"values" = full state, "updates" = just changes, "custom" = custom events from tools



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What is LangSmith and how is it used?

LangSmith is LangChain's debugging and monitoring platform. It shows you exactly what's happening inside your AI app!



Tracing: See every step of your chain/agent



Debugging: Find why something went wrong



Evaluation: Test your app's quality

```
import os
os.environ["LANGSMITH_TRACING"] = "true"
os.environ["LANGSMITH_API_KEY"] = "your-key"
# That's it! All runs are now traced
```

PYTHON



How do you handle errors in LangChain agents?

Agents can fail! Use **error handling** to retry or recover. Always return helpful error messages so AI can try alternatives.

PYTHON

```
# Handle errors in the tool itself
@tool
def safe_search(query: str):
    try:
        return do_search(query)
    except Exception as e:
        return f"Search failed: {e}"

# Or use ToolStrategy for auto-retry
from langchain.tools import ToolStrategy
strategy = ToolStrategy(
    schema=MyTool,
    handle_errors=(ValueError, TypeError)
)
```



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What are best practices for RAG optimization?

Better RAG = Better answers! Here's how to optimize:

-  **Chunk size:** Experiment with different sizes (500-2000 chars)
-  **Overlap:** 10-20% overlap prevents lost context
-  **Hybrid search:** Combine vector + keyword search
-  **Reranking:** Re-order results by relevance
-  **Metadata:** Filter by date, source, category



How do you implement dynamic prompts?

Dynamic prompts change based on context, user, or conversation state. Great for personalization!

PYTHON

```
from langchain.agents.middleware import dynamic_prompt,
ModelRequest
```

```
@dynamic_prompt
```

```
def smart_prompt(request: ModelRequest) → str:
```

```
    # Check conversation length
```

```
    msg_count = len(request.messages)
```

```
    base = "You are a helpful assistant."
```

```
    if msg_count > 10:
```

```
        base += "\nBe extra concise."
```

```
    return base
```

```
agent = create_agent(model, middleware=[smart_prompt])
```



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What is human-in-the-loop in LangChain?

Human-in-the-loop = Pause the AI and ask a human for approval before taking critical actions.

USE CASES

Sending emails, making purchases, deleting data, or any action that can't be undone!



interrupt_before: Pause before specific tools run



interrupt_after: Pause after tools, review results



Edit state: Human can modify before resuming



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How do you deploy LangChain apps to production?

Production-ready LangChain apps need these considerations:



API Keys: Use environment variables, never hardcode



Caching: Cache embeddings and responses to save costs



Rate Limiting: Protect against abuse



Monitoring: Use LangSmith for observability



LangServe: Deploy chains as REST APIs easily



*Tap that follow button
and stay in the loop!*



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